

Pediatric traumatic hemorrhagic shock consensus conference recommendations

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ABSTRACT: Hemorrhagic shock in pediatric trauma patients remains a challenging yet preventable cause of death. There is little high-quality evidence available to guide specific aspects of hemorrhage control and specific resuscitation practices in this population. We sought to generate clinical recommendations, expert consensus, and good practice statements to aid providers in care for these difficult patients. The Pediatric Traumatic Hemorrhagic Shock Consensus Conference process included systematic reviews related to six subtopics and one consensus meeting. A panel of 16 consensus multidisciplinary committee members evaluated the literature related to 6 specific topics: (1) blood products and fluid resuscitation for hemostatic resuscitation, (2) utilization of prehospital blood products, (3) use of hemostatic adjuncts, (4) tourniquet use, (5) prehospital airway and blood pressure management, and (6) conventional coagulation tests or thromboelastography-guided resuscitation. A total of 21 recommendations are detailed in this article: 2 clinical recommendations, 14 expert consensus statements, and 5 good practice statements. The statement, the panel's voting outcome, and the rationale for each statement intend to give pediatric trauma providers the latest evidence and guidance to care for pediatric trauma patients experiencing hemorrhagic shock. With a broad multidisciplinary representation, the Pediatric Traumatic Hemorrhagic Shock Consensus Conference systematically evaluated the literature and developed clinical recommendations, expert consensus, and good practice statements concerning topics in traumatically injured pediatric patients with hemorrhagic shock. (*J Trauma Acute Care Surg.* 2023;94:S2–S10. Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.)

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Trauma is the leading cause of pediatric mortality, potential years of life lost, and a significant medical cost in the developed world.^{1,2} Thirty-day mortality in children with traumatic hemorrhagic shock is 36% to 50% compared with 20% to 25% reported in adults.^{3,4} An estimated 1,000 to 2,000 preventable traumatic deaths in children per year after injury occur in

the United States because of inadequate or delayed care.⁵ Recent retrospective and prospective observational studies indicate that transfusion strategies (limiting crystalloid, appropriate transfusion ratios, and use of whole blood [WB]) and intravenous hemostatic adjunct therapies can reduce morbidity and mortality in children with traumatic hemorrhagic shock.^{6–9} For this

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reason, hemostatic strategies represent a key target for standardization and subsequent investigation. However, because of the lack of high-quality clinical trials to guide practice, hemostatic resuscitation strategies and the utilization of global assays of hemostasis for goal-directed therapy are highly variable across sites.^{10,11} There is also a lack of agreement upon critical definitions of pediatric massive transfusion, definition of coagulopathy, and critical common data elements required for clinical trial design. To address these challenges, we organized a multidisciplinary team of experts and key stakeholders to (1) develop consensus statements on best practice in resuscitation strategies for pediatric trauma patients experiencing hemorrhagic shock based on the current literature; (2) create a strategy, in collaboration with implementation experts, for adaptive dissemination and implementation into clinical and research environments; and (3) develop future research priorities for studying resuscitation practices for pediatric traumatic hemorrhagic shock and foster collaboration in pursuit of improved clinical care for these patients.

PATIENTS AND METHODS

The methodology for the Pediatric Traumatic Hemorrhagic Shock Consensus Conference follows the standards set by the Institute of Medicine for developing comprehensive evidence-based recommendations. When evidence was lacking, the panel developed expert consensus statements and good practice statements for decision making in pediatric trauma patients experiencing hemorrhagic shock. The complete methods are in Appendix A, <http://links.lww.com/TA/C726>, and the good practice statements in Appendix B, <http://links.lww.com/TA/C727>.

Clinical Recommendations (n = 2) and Expert Consensus Statements (n = 14)

1. Blood Products and Fluid Resuscitation in Pediatric Traumatic Hemorrhagic Shock

1.1. In traumatically injured children in hemorrhagic shock, we suggest prioritizing the use of blood products over the use of crystalloids for resuscitation (consensus panel expertise; 100% agreement [n = 16]; median, 9; interquartile range [IQR], 8–9).

Rationale

During and after the Vietnam War, crystalloids and colloids replaced blood as the primary initial resuscitative solution for hemorrhagic shock. This change was in part due to logistic difficulties of using blood and the infectious disease risks of blood products. The shift to crystalloids was supported by research performed by Carrico et al.¹² indicating that the interstitial compartment or “third space” needed to be resuscitated with 1 to 2 L of crystalloids to perfuse the tissues. This research further recommended that the transfusion of WB would only be indicated if hemodynamic instability persisted following administration of crystalloids.¹² To the detriment of patients with severe bleeding, misinterpretation of these data contributed to overuse of crystalloids before any blood product was administered. When used in practice, this approach could result in dilutional coagulopathy and severe interstitial edema. A shift from early blood product administration to a preference for early crystalloid and colloid use led to an increase in acute respiratory distress

syndrome, abdominal compartment syndrome, multiorgan failure, and anasarca in intensive care units.¹³

These outcomes were predicted by Shoemaker in 1976¹⁴ when he challenged the notion that the interstitial compartment required resuscitation. He instead emphasized the need for WB to treat significant bleeding when the hematocrit fell below 30%. The overuse of crystalloids continued despite a call for changes in practice recommended by Moore and Shires¹⁴ as early as 1967. In an editorial, Moore and Shires¹⁴ state, “blood should still be replaced during major operative surgery as it is lost. The use of balanced salt solutions appears to be a physiological adjunct to surgical trauma, not a substitute for blood.” Subsequent research has demonstrated that a crystalloid-based resuscitation strategy leads to higher inflammation and vascular permeability compared with WB.¹⁴

Data supporting the use of blood products instead of crystalloids in children include a multicenter prospective observational study performed from April 2018 to September 2019.¹⁵ In this study of 712 children from 24 trauma centers, each crystalloid bolus after the first was incrementally associated with an increased odds of more mechanical ventilation and intensive care unit days and longer hospital stays. A longer time to initiate the first transfusion was also associated with more mechanical ventilation days. In another retrospective study of 512 children admitted to combat support hospitals between 2007 and 2016, the authors evaluated the association of high crystalloid use (>40 mL/kg in 24 hours) for patients receiving high or low plasma/red blood cell (RBC) ratios (1:2 threshold).¹⁶ In children who received a high ratio of plasma/RBCs and low crystalloid volume, there was an independent association with improved survival (odds ratio [OR], 3.42; 95% confidence interval [CI], 1.04–11.24). In contrast, children with a high ratio of plasma/RBCs with high crystalloid volumes was not associated with improved survival (OR, 0.61; 95% CI, 0.28–1.29). The authors concluded that the high crystalloid volume negated the potential survival benefit of high plasma/RBC ratios in children with severe traumatic bleeding.

A secondary analysis of the Prehospital Air Medical Plasma trial showed similar findings. This study of injured adults compared four groups of patients based on prehospital resuscitation with packed red blood cells (PRBC) and plasma, plasma only, PRBC only, and crystalloid only. For patients with severe traumatic bleeding, they found that the exclusive use of crystalloids worsened survival compared with the partial use of blood products in the prehospital resuscitation phase.¹⁷ A second randomized control trial in adults with severe traumatic injury showed that the use of crystalloids in the prehospital phase of resuscitation decreased the risk of survival compared with not using any crystalloids or blood products.¹⁸

Pediatric and adult data support the association of crystalloids with worse outcomes and the increased and early use of blood products (WB or balanced approach) with improved survival. These data support our recommendation suggesting the prioritization of blood products over crystalloids for the resuscitation of traumatic hemorrhagic shock in children.

1.2. In traumatically injured children in hemorrhagic shock, the use of low titer (≤ 200 immunoglobulin G) group O WB might be considered if available over individual blood

components (RBC, plasma, and platelets) for resuscitation (clinical recommendation; conditional recommendation; very low certainty of evidence; 94% agreement [$n = 16$]; median, 8.5; IQR, 7.75–9).

Rationale

When compared with individual blood components in a balanced ratio, low-titer group O whole blood (LTOWB) is more potent because of less anticoagulants, less preservatives increasing hemoglobin, platelet concentrations, and coagulation factors.¹⁹ In addition, storage of platelets at 4°C may also make LTOWB more hemostatic. In addition, WB causes less dilutional coagulopathy and less hypocalcemia because of the increased citrate when transfusing multiple blood components.²⁰ Because of the exclusive use of a group O product, WB may be safer than use of components by preventing transfusion of incompatible ABO unit(s) that may lead to a fatal hemolytic reaction due to human error.²¹ In addition, the storage of WB at 4°C reduces the risk of bacterial contamination that exists with the use of platelets stored at room temperature.²² Finally, WB has logistic advantages because of the need to only dispense one product from the blood bank and administer one product at the bedside instead of dispensing and administering individual blood components.

Using LTOWB instead of individual blood components in children more rapidly provides RBCs, plasma, and platelets to children with severe traumatic bleeding; more effectively resolves shock and coagulopathy²³; is associated with less total amount of blood products administered and mechanical ventilation days²⁴; and is independently associated with increased 72-hour (OR, 0.23; 95% CI, 0.08–0.70) and 28-day mortality (OR, 0.41; 95% CI, 0.23–0.98) in a single-center retrospective study.⁸

Adult data indicate that the use of LTOWB compared with individual blood components is independently associated with improved 24-hour and 28-day survival and is also associated with less (40–60%) total blood products administered.^{25–27} Data from studies performed on children and adults suggest no increased risk of hemolysis in non-group O recipients and no increase in any other adverse outcomes to include organ failure.^{28–30}

1.3. In traumatically injured children in hemorrhagic shock, when using blood component resuscitation, we suggest targeting high plasma/RBC ratios (1:1) to minimize the plasma deficit (consensus panel expertise; 100% agreement [$n = 16$]; median, 9; IQR, 8–9).

Rationale

Children with life-threatening hemorrhage from traumatic injuries develop shock due to hypoperfusion. This hypoperfusion causes endothelial injury with loss of the glycocalyx and increased endothelial permeability. The endothelial injury leads to inflammation, which can further exacerbate endothelial injury and lead to the production of increased activated protein C and tissue plasminogen activator directly causing trauma induced coagulopathy. While this pathophysiology has been mainly described in adults, there are data indicating that it may also occur in children.^{31–33} Differences in hemostasis mechanisms between children and adults will require additional study to establish whether these mechanisms are also observed in children.³⁴ To

address the shock, endothelial, immune, and hemostatic dysfunction, a balanced resuscitation with plasma and RBCs may be optimal because plasma may repair the endothelium and mitigate capillary leak and improve intravascular volume. Plasma may also improve hemostasis by providing coagulation factors for patients who have developed a consumptive coagulopathy. These potential effects of plasma may improve intravascular volume and reduce extravascular edema and bleeding, which can all lead to improved oxygen delivery.

A 2-year retrospective review of the pediatric trauma quality improvement program supports that the increased plasma/RBC ratios minimize plasma deficit. In this study of more than 500 massively transfused children, the plasma/RBC ratio, as a continuous variable, was associated with improved 24-hour mortality (OR, 0.47; 95% CI, 0.28–0.80). In this same study, a plasma/RBC ratio >1:1 was also independently associated with reduced 24-hour mortality (OR, 0.48; 95% CI, 0.26–0.88). In a multi-institutional prospective observational study that included 191 children with traumatic injury and life-threatening hemorrhage, there was an independent association with a plasma/RBC ratio of >1:2 with improved 24-hour survival (OR, 0.36; 95% CI, 0.13–0.99). When the plasma deficit (RBC mL/kg – plasma mL/kg) was analyzed, an increased deficit was also associated with increased 24-hour mortality (OR, 1.2; 95% CI, 1.05–1.3).⁴ Additional publications that do not report an association of increased plasma/RBC ratios with improved outcomes have significant limitations to include the lack of adjusted analyses and small sample size or single-center studies.^{27,35,36} Adult data supporting the use of high ratios targeting 1:1 include the PROPPR (Pragmatic, Randomized Optimal Platelet and Plasma Ratios) trial that reported reduced death from bleeding at 24 hours and improved time to cessation of bleeding.³⁷ The PAMPer (Prehospital Air Medical Plasma) trial also reported improved survival when plasma was used early in the prehospital phase of resuscitation.³⁸

1.4. In traumatically injured children in hemorrhagic shock, when using blood component resuscitation, targeting a high platelet to RBC weight-based ratio of 1:1 to minimize the platelet deficit is suggested (consensus panel expertise; 100% agreement [$n = 16$]; median, 8; IQR, 7–9).

Rationale

Platelet dysfunction occurs early in traumatic blood failure in adults and the transfusion of platelets may reverse it.³⁷ The importance of the platelets within hemostasis is another reason to hypothesize that the use of platelets may reduce bleeding and improve outcomes. In addition, in vitro and animal evidence suggest that platelet transfusion may repair the injured endothelium, which could also improve outcomes.^{39,40}

Data supporting the early use of platelets in children were also published in the aforementioned multi-institutional prospective observational study in children with life-threatening bleeding. In this study, a platelet deficit (RBC mL/kg – platelet mL/kg) was independently associated with increased mortality at 24 hours (OR, 1.1; 95% CI, 1.05–1.2), but there was no survival advantage associated with the platelet/RBC ratio.⁴ A plausible explanation for this may be that the deficit more accurately reflects the lack of balance between platelets and RBCs more than a ratio, which does not incorporate the magnitude of the imbalance. Additional

adult data supporting the early use of platelets, a secondary analysis of Pragmatic, Randomized Optimal Platelet and Plasma Ratios trial focusing on platelet transfusions, demonstrated improved survival in these patients who achieved a more balanced resuscitation.⁴¹

2. Prehospital Blood Products Use in Pediatric Traumatic Hemorrhagic Shock

2.1. In traumatically injured children in hemorrhagic shock, it is reasonable to consider prehospital transfusion by out-of-hospital emergency medical service (EMS) for injured children based on product availability and clinical judgment (consensus panel expertise; 100% agreement [n = 16]; median, 8; IQR, 8–8.25).

Definition for Purpose of This Recommendation

Out-of-hospital EMS is defined as all medical care for a trauma patient that occurs in a non-health care setting, including scene response and interhospital transfer.

Rationale

Death due to bleeding most often occurs early after injury.⁴² Delays in time to transfusion are associated with increased mortality and increased time to hemostasis.⁴³ The rationale for prehospital transfusion is to shift resuscitation earlier to address blood loss, mitigate shock and coagulopathy, and ultimately reduce mortality and morbidity for injured patients.

Little high-quality evidence has defined the optimal approach to fluid resuscitation and transfusion by out-of-hospital Emergency Medical Services (EMS) in children with hemorrhagic shock. The management of acute life-threatening bleeding has been studied almost exclusively in adult trauma, where early blood product use by out-of-hospital EMS providers has become an accepted and common practice. In military^{44–46} and civilian^{38,47} adult cohorts, the use of blood products as the first resuscitative fluid has been associated with improved survival. No studies to date have reported survival benefits of prehospital blood product transfusion in children.

The transfusion of blood products by out-of-hospital EMS providers is feasible and safe.^{48,49} Data show that the LTOWB,⁴⁹ red blood cells, and plasma⁴⁸ under pediatric dosing protocols is feasible, acceptable to providers, and not associated with adverse transfusion-related events.⁵⁰ Only two studies have compared outcomes between matched pediatric cohorts who received blood products prehospital versus in-hospital. These studies are limited by sample size and moderate bias because of outcome measurement and confounding.^{50,51} Although the studies were not adequately powered to detect a difference in mortality, authors did report that the prehospital transfusion groups did receive less in-flight crystalloid and had higher admission fibrinogen levels and faster time to normalization of the international normalized ratio.⁵¹ This recommendation has to take into account the ability of blood products and comfort level of the practice of regional EMSs. However, the previous literature suggests that there is evidence that this can be organized and safely used in pediatric trauma patients with hemorrhagic shock.

3. Use of Tranexamic Acid and Other Hemostatic Adjuncts in Pediatric Traumatic Hemorrhagic Shock

3.1. In traumatically injured children with hemorrhagic shock, the empiric use of tranexamic acid within 3 hours of injury

might be considered (clinical recommendation; conditional recommendation; very low certainty of evidence; 80% agreement [n = 15]; median, 7; IQR, 7–8).

Rationale

Fibrinolytic dysregulation is an important mechanism in traumatic coagulopathy that consists of a spectrum ranging from excessive breakdown (hyperfibrinolysis) to physiologic fibrinolysis and to fibrinolytic shutdown. Injury to the vascular bed with endothelial disruption initiates a complex interaction between the endothelium, platelets, and coagulation factors ultimately resulting in thrombin generation and cross-linking of fibrin monomers. This interaction results in the formation of a mature clot, seals the area of vascular injury, and leads to hemostasis. Fibrinolysis is a physiologic process occurring in parallel with fibrin cross-linking to prevent the extension of this process beyond the injury. During injury, this process can turn from physiologic to pathologic and then to increased or reduced fibrinolysis, which has been associated with high mortality.⁵² This increases the importance of timely identification and treatment of fibrinolytic dysregulation in trauma patients. Tranexamic acid (TXA), an antifibrinolytic agent, has been extensively studied in adult trauma trials in an effort to improve outcomes in patients with hyperfibrinolysis.^{53–55} We did not identify any randomized clinical trials evaluating TXA in severely injured children. There were five observational studies (four retrospective and one prospective study) that compared empiric TXA use versus no TXA, controlling for confounders. Three of the studies were in civilian settings and two in combat settings. One combat associated paper evaluated a registry of 766 children of 9% received TXA.⁷ Using propensity score match analysis, they found that TXA was independently associated with a lower odds of mortality (OR, 0.87; 95% CI, 0.85–0.89). The second combat study also evaluated a registry of children receiving massive transfusion and demonstrated that TXA use was associated with a lower odds of mortality (OR, 0.35; 95% CI, 0.12–0.99).⁵⁶ A prospective observational study of children with life-threatening bleeding showed that antifibrinolytic agents (TXA or aminocaproic acid) were associated with decreased 6- and 24-hour mortality (6 hours: adjusted OR, 0.29; 95% CI, 0.09–0.93; 24 hours: adjusted OR, 0.45; 95% CI, 0.21–0.98) compared with those who received no antifibrinolytic agent.⁵⁷ Two retrospective studies did not find an association between TXA use and improved outcomes. One study evaluated a large administrative trauma database in Japan with a propensity matched analysis of children who did and did not receive TXA.⁵⁸ They found no difference in mortality between groups. However, based on a low mean Injury Severity Score (ISS) of 10, they may not have identified pediatric population most at risk for hemorrhagic shock. The second retrospective study evaluated massively transfused injured children and compared patients who received TXA with those who did not.⁵⁹ They found no mortality difference between groups, but this study had a small sample size (n = 48).

There is strong evidence for empiric TXA in severely injured adults. The CRASH-2 (Clinical Randomization of an Antifibrinolytic in Significant Hemorrhage) randomized control trial randomized more than 20,000 injured adults.⁶⁰ Compared with placebo, TXA decreased death from bleeding by one third if given within

3 hours from injury. Several additional prospective adult randomized studies also have found a strong correlation of TXA administration with survival if given soon after injury, including when given to head injured patients.^{54,61,62}

3.2. In traumatically Injured children with hemorrhagic shock, we recommend the use of TXA over aminocaproic acid when an antifibrinolytic agent is being considered (consensus panel expertise; 93% agreement [n = 15]; median, 8; IQR, 7–9).

Rationale

No high-quality studies comparing TXA with aminocaproic acid have been published in a pediatric trauma population. However, a much broader body of adult and pediatric trauma and surgical literature exists to show the efficacy and safety of TXA than for aminocaproic acid.^{7,53,55,56,63–65} In addition, some data suggest an association between acute kidney injury and aminocaproic acid use in all children with life-threatening bleeding.⁵⁷

3.3. In traumatically injured children with hemorrhagic shock, there is insufficient evidence for the empiric use of prothrombin concentrate (consensus panel expertise; 93% agreement [n = 15]; median, 9; IQR, 7–9).

Rationale

Prothrombin complex concentrate (PCC) contains either three or four of the vitamin K-dependent coagulation factors (II, IX, X, and sometimes VII) and may be an alternative approach, to FFP, for correction of coagulopathy.⁶⁶ Although the literature is limited, adult literature in both trauma and operative settings supports the administration of PCC to bleeding patients to reverse coagulopathy.^{67,68} No high-quality studies evaluating the use PCC in a pediatric trauma population were identified. In a small retrospective study of a mixed pediatric population with only half experiencing trauma, PCC was found to improve international normalized ratio but had no effect on outcomes.⁶⁹ While PCC is commonly used as a reversal agent in bleeding associated with vitamin K antagonists or direct oral anticoagulants, there is little clinical or physiologic evidence that PCC has any benefit as an empiric treatment for traumatically injured children in hemorrhagic shock.

3.4. In traumatically injured children with hemorrhagic shock, there is insufficient evidence that the use of viscoelastic monitoring (VEM) to guide antifibrinolytic therapy improves patient outcomes (consensus panel expertise; 94% agreement [n = 16]; median, 7; IQR, 7–8.25).

Rationale

Several studies have identified that children with hyperfibrinolysis have a higher mortality after injury.^{70,71} Given this observation and the understanding of the antifibrinolytic effects of TXA, researchers would suggest that those children with hyperfibrinolysis should be identified and targeted for TXA.⁷² Some pediatric trauma centers may use VEM to evaluate trauma patients for coagulopathy and fibrinolysis in addition to conventional coagulation testing. However, the selective use of TXA based on VEM has not been studied in children, and the limited adult studies for rapid-thromboelastography-guided TXA administration have failed to show a clear benefit.⁷³ Until additional literature evaluates the specific utilization of VEM before

and following administration of TXA in pediatric patients with different postinjury fibrinolysis subtypes, we cannot recommend use of VEM in this setting.

3.5. In traumatically injured children with hemorrhagic shock, there is insufficient evidence for the empiric use of fibrinogen supplementation (cryoprecipitate or fibrinogen concentrates; consensus panel expertise; 87% agreement [n = 15]; median, 7; IQR, 7–8).

Rationale

No studies supporting the empiric use of fibrinogen or cryoprecipitate in pediatric hemorrhagic trauma were identified. There have been several adult studies investigating the use of fibrinogen supplementation; however, a meta-analysis of these studies failed to show a mortality benefit.⁷⁴

3.6. In traumatically injured children with hemorrhagic shock, we suggest the replacement of fibrinogen in the setting of hypofibrinogenemia (consensus panel expertise; 87% agreement [n = 16]; median, 8; IQR, 8–9).

Rationale

It is established that fibrinogen is an essential substrate for clot formation and that low fibrinogen can lead to severe bleeding whether congenital or acquired.³¹ In the setting of trauma, low fibrinogen is essentially due to consumption. Most trauma centers will screen patients for hypofibrinogenemia, particularly during a massive transfusion, and correct if the fibrinogen level is below 150 mg/dL.⁷⁵ Low fibrinogen levels can be corrected either by the administration of fibrinogen concentrate or cryoprecipitate. Viscoelastic monitoring has been used to identify low fibrinogen, based on prolonged R time, low α angle, or low maximum amplitude. Rotational thromboelastometry can also be used. A pediatric study is underway studying early fibrinogen concentrate in pediatric patients presenting with a FIBTEM (extrinsically activated thromboelastometric test with cytochalasin D) A5 of ≤ 10 .⁷⁶ This study may provide us additional guidance for using of fibrinogen in pediatric trauma patients.

4. Use of Tourniquets in Pediatric Traumatic Hemorrhagic Shock

4.1. In traumatically injured children with exsanguinating extremity hemorrhage, we recommend the use of commercially available tourniquets by individuals with training (consensus panel expertise; 88% agreement [n = 16]; median, 9; IQR, 7.75–9).

Rationale

In the past, the use of tourniquets for exsanguinating extremity hemorrhage has had mixed impact on outcomes, mainly related to inappropriate applications by untrained responders. In earlier military conflicts, challenges with delays in reaching definitive care led to complications including the risk of limb loss.^{77,78} Recent adult military experience with quicker transport times and enhanced training among soldiers/first responders showed a clear survival advantage among those with tourniquet use.⁷⁹ These data along with a dramatic increase in mass casualty events have influenced a paradigm shift among first responders in civilian settings toward tourniquet use.^{80–83} Education of first responders has been advanced by a committed effort

by the American College of Surgeons “Stop the Bleed” program. Concerns remain regarding the applicability of combat adult practice with relation to children in civilian settings.

In these elective settings, tourniquet complications were found to occur in 0.4% to 1.4% of patients, with most injuries involving soft tissue or nerve damage.⁸⁴

Our review of available evidence for the use of tourniquets in exsanguinating extremity hemorrhage in children yielded six manuscripts: four from a military setting^{85–88} and two evaluating a civilian experience.^{89,90} These studies showed decreased crystalloid administration, decreased transfusion requirements, and a survival advantage for children treated with tourniquets, particularly when applied before the onset of shock. There were no significant complications from tourniquet use. In the studies evaluated, challenges still exist with tourniquet use including tourniquet overuse, improper tourniquet application, and a need for more standardized training.⁸⁹

5. Prehospital Intubation and Blood Pressure Management in Pediatric Traumatic Hemorrhagic Shock

5.1. In traumatically injured children with hemorrhagic shock, we suggest against a permissive hypotension strategy and suggest resuscitation goals that optimize end organ perfusion and adequate oxygen delivery (consensus panel expertise; 93% agreement [n = 15]; median, 7; IQR, 7–8.5).

Rationale

Although some reports in the adult trauma literature suggest a potential benefit of permissive hypotension,^{91,92} this has become more controversial.^{93,94} In pediatric trauma patients, there was insufficient evidence to evaluate this strategy. However, there is a widely accepted paradigm that children have a significant ability to compensate until very late stages of hemorrhagic shock.⁹⁵ In Advanced Trauma Life Support, the American College of Surgeons Committee on Trauma reports that a child has a low but normal blood pressure starting after 30% blood volume loss and experiences hypotension only after 45% blood volume loss.⁹⁶ Furthermore, a rigorous definition of permissive hypotension in pediatric patients could not be identified in the literature. Finally, in addition, injured children with hypotension may have a concomitant head injury for which permissive hypotension would not be a recommended strategy.

6. Resuscitation and Use of Hemostatic Monitoring for Pediatric Traumatic Hemorrhagic Shock

6.1. In traumatically injured children with hemorrhagic shock, we suggest an initial empiric resuscitation approach using massive transfusion protocols (MTPs) and balanced blood product administration (consensus panel expertise; 94% agreement [n = 16]; median, 9; IQR, 7–9).

Rationale

In severely injured adult trauma patients, empiric hemostatic resuscitation using balanced blood products or WB as part of an MTP is standard of care.^{97,98} Although there have been many studies evaluating massively transfused children, a paucity of studies has critically examined the benefits of MTP implementation and use in children.⁹⁹ Three single-center, retrospective

studies specifically evaluated the impact of MTPs on outcomes. A 2011 study compared 33 pre-MTP to 22 post-MTP children from a 3-year period. The pre-MTP and post-MTP groups included 31 and 16 trauma patients, respectively. Overall, there was no difference in mortality among the group even though the post-MTP implementation patients were more severely injured as assessed by ISS. The post-MTP group received more blood products and experienced less thromboembolic complications.¹⁰⁰ A second 2011 study compared 53 MTP patients with 49 non-MTP patients over a 2-year period. There were no differences in baseline characteristics including ISS. The MTP group received plasma earlier and received a more balanced resuscitation, but no difference in mortality was observed.¹⁰¹ Finally, a 2016 study compared 125 pre-MTP with 115 post-MTP implementation patients over a 10-year period. Despite the post-MTP cohort being more severely injured, there was no difference in mortality. Post-MTP patients received plasma and platelet transfusions earlier and in a more balanced ratio with red blood cells.¹⁰² These studies suggest that MTPs are safe and may provide benefit in the resuscitation of severely injured pediatric trauma patients in hemorrhagic shock.

6.2. In traumatically injured children with hemorrhagic shock, following empiric initial resuscitation, a goal-directed resuscitation strategy to optimize hemostasis and correct coagulopathy is suggested (consensus panel expertise; 100% agreement [n = 15]; median, 8; IQR, 7.5–9).

Rationale

A goal-directed hemostatic resuscitation strategy to optimize hemostasis and correct coagulopathy is overwhelmingly recommended and supported by high-quality evidence in traumatically injured adult patients.^{97,98} We were unable to identify a pediatric-focused study that directly compared a goal-directed approach to a non-goal-directed approach following initial resuscitation with an MTP. However, eight studies were identified describing the use of measures of hemostasis in the resuscitation of injured children. These studies illustrate the importance of assessing abnormalities in coagulation during the trauma resuscitation and early hospital course. During the resuscitation process, tachycardia, hypotension, and clinical and laboratory markers of impaired end organ perfusion begin to resolve when hemodynamic stability is established. Continued administration of blood products as part of an MTP for hemodynamic resuscitation may be inefficient, costly, and potentially harmful. A transition to a more targeted approach for blood product administration to correct specific coagulation parameters is reasonable.

6.3. In traumatically injured children with hemorrhagic shock, adjunctive VEM when available is suggested (consensus panel expertise; 100% agreement [n = 15]; median, 8; IQR, 7–9).

Rationale

In children, studies have shown that abnormalities in VEM parameters correlate with conventional coagulation tests (CCTs) abnormalities, injury severity, and mortality.^{103,104}

Studies in children have shown that, compared with CCT, VEM parameters were more closely correlated with needing a lifesaving intervention and mortality.^{103,104} A retrospective, multicenter study compared outcomes in children before and after implementing a VEM-based resuscitation strategy. Using propensity score matching, the authors found that patients who received a VEM-based resuscitation strategy had more timely results, received less 24-hour red blood cell and plasma volumes, were less coagulopathic after 24 hours, and had less total hospital days. There was no difference in mortality.¹⁰⁵

Viscoelastic monitoring, including both thromboelastography and rotational thromboelastometry, provides accurate and comprehensive depictions of traumatically injured adults coagulation profiles in hemorrhagic shock.¹⁰⁶ Viscoelastic monitoring provides more timely results compared with CCT.^{107,108} Randomized prospective studies in severely injured adults receiving an MTP-based resuscitation for hemorrhagic shock showed reduced morbidity and mortality in patients who received a VEM-based resuscitation.¹⁰⁸ We believe that utilization of VEM may aid in resuscitation of children in hemorrhagic shock, but as noted in recommendation 3.4, we do not believe that there are sufficient data in children to use VEM for monitoring fibrinolysis or directing administration of TXA.

CONCLUSION

The Pediatric Traumatic Hemorrhagic Shock Consensus Conference aimed to provide guidance and recommendations for the resuscitation of traumatically injured children in hemorrhagic shock. Despite the lack of or low quality of the literature in the pediatric population, our expert panel has reached agreement on clinical recommendations, consensus, and good practice statements to guide clinicians in care for these patients.

AUTHORSHIP

R.T.R., J.R.E., M.M.B., and P.J.A. reviewed abstracts and full-text articles for the topic literature reviews. R.T.R., M.M.B., M.A.B., D.K.N., P.A.S., A.M.V., C.L., and M.J. drafted portions of the manuscript, and R.T.R. synthesized the manuscript. All authors reviewed the reviewed, edited, and approved the manuscript before final submission.

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DISCLOSURE

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